Mehmet Emin Meydanoğlu

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Experience

ITU Erdal Aksoy Industrial Robotics and Artificial Intelligence Laboratory
Istanbul, Turkiye

Undergraduate Researcher — Sep 2026 – Present

Ongoing research on world action models and training policies that generalize with no task-specific demo.

Hyperever
Istanbul, Turkiye

Robotics Software Engineer Intern — Jul – Aug 2026

1. ***Quadruped Blind Rough Terrain Locomotion*** Developed proprioceptive latent-context training methods for quadruped locomotion, inspired by algorithms like RMA, DreamWaQ, and HIM. Tested whether policies could adapt to the environment using latent context inferred from proprioceptive history. Used explicit system identification as the baseline. Transferred a policy trained in Genesis to MuJoCo, where it continued to successfully traverse rough terrain.
2. ***VLM Operator Assistance*** Built a prototype that turned live footage from the robot's front camera into short safety notes for the operator. Uses open-weight edge models like Gemma and Qwen to identify people, obstacles, and terrain conditions and suggest a safe next move. Benchmarked models and frame sampling settings for latency and memory use.

ITU AUV Team
Istanbul, Turkiye

Co-Lead of Autonomous Software — Sep 2025 – Mar 2026

Team Member — Sep 2024 – Aug 2026

Building autonomous underwater vehicles (AUVs) for international competitions and academic research. The team placed **third** worldwide at RoboSub 2025, **second** worldwide at RoboSub 2026, and **first** nationally at Teknofest 2026. github.com/itu-auv/auv-software

1. ***Learning-based autonomy*** Experimented with imitation and reinforcement learning for 6-DOF vehicle control and underwater manipulation with robotic grippers, using vision transformer (ViT) image embeddings as policy observations.
2. ***Automated image labeling*** Developed workflows to automate dataset labeling for object detection and segmentation. This work evolved into agent-seg.
3. ***Classic Autonomy*** Implemented a custom Pure Pursuit–style controller for trajectory tracking, worked on EKF-based odometry and cascaded PID control, and designed autonomous tasks using state machines.

Keywords Reinforcement learning for control, ViT, YOLO, Depth Anything, ROS, visual SLAM

ITU Racing Team
Istanbul, Turkiye

Autonomous Software Team Member — Dec 2023 – Aug 2024

Worked on early-stage autonomous driving software experiments in simulation. Small projects in SLAM, path planning and control for a student formula autonomous vehicle project.

Projects

agent-seg github.com/eminmeydanoglu/agentseg

Tools for agents to label video. Agent will use SAM to track any objects across frames, check and correct labels, and export to any format easily.

auv-rl github.com/eminmeydanoglu/auv-rl

Codebase to train reinforcement learning control policies for autonomous underwater vehicles (AUVs).

self-taught-ViT github.com/eminmeydanoglu/self-taught-vit

Proof-of-concept self-supervised Vision Transformer trained with masked image reconstruction, proving even a very small ViT can learn semantic representations without any labels.

Education

Istanbul Technical University — **Istanbul, Turkiye**

B.S. Mathematical Engineering *(GPA: 3.24/4.00)* — *Sep 2025 – Present*

B.S. Control and Automation Engineering *(Transferred)* — *Sep 2023 – Sep 2025*

Extracurriculars

Philosophy and science communication: Produce YouTube videos on philosophy of mind, AI, intelligence, Stoicism, physics, and religion for a general audience. youtube.com/eminmeydanoglu

Essays: Write about philosophy, psychology, human nature, meaning, and self-cultivation. meydanoglu.com/writings

Sports: Tennis, running, and cycling.